

2026-06-10-impl-batch3-review

Wave-3 Implementation Batch — Independent Code Review (§11.4.125 / §11.4.142)

Date: 2026-06-10 **Reviewer:** Independent code-review subagent (read-only; no edits/commits/push) **Scope:** UNCOMMITTED wave-3 code — SP1-funnel (helix_code) + SP2 catalog (submodules/helix_agent) **Verdict:** GO

1. Batch under review

SP1-funnel (helix_code, all builds/tests from helix_code/): - internal/llm/keyrecognition.go — new PresentProviderNames() map[string]bool (string-keyed bridge over PresentProviders()). - cmd/server/main.go — new loadAPIKeysAtStartup() wired into main() BEFORE config.Get(). - internal/server/handlers.go — listLLMModels + listLLMProviders switched GetVerifiedModels(ctx) → GetWorkingModels(ctx, llm.PresentProviderNames()). - cmd/cli/main.go — handleListModels production call site switched to llm.PresentProviderNames(). - New tests: cmd/server/loadapikeys_wiring_test.go, internal/server/llm_working_funnel_test.go, cmd/cli/listmodels_funnel_source_test.go, internal/llm/keyrecognition_test.go edits.

SP2 (submodules/helix_agent): - New internal/catalog/{catalog,registry_adapter,handler,defaults,catalog_test,handler_test}.go. - internal/router/router.go — registered protected.GET("/catalog", ...) at line 785 + import reorder.

2. VERIFY results (commands run)

2.1 Build + vet — PASS

```
# SP1 (helix_code)
go build ./internal/llm/... ./internal/server/... ./cmd/server/... ./cmd/c
go vet   ./internal/llm/... ./internal/server/... ./cmd/server/... ./cmd/c
# SP2 (submodules/helix_agent)
go build ./internal/catalog/... ./internal/router/...
go vet   ./internal/catalog/...
```

2.2 RED→GREEN real — PASS

SP1 funnel/wiring tests (GREEN, explicit RED_MODE=0 — tests default unset/1 to RED):

```

RED_MODE=0 TestServerLoadAPIKeys_WiredAtStartup (cmd/server)
RED_MODE=0 TestCLI_ModelListing_UsesCommittedKeyRecognitionSource (cmd/cli)
RED_MODE=0 TestPresentProviderNames_StringKeyedFunnelInput (internal/llm)
RED_MODE=0 TestServerListLLMModels_WorkingFunnelEndToEnd (internal/server)
RED_MODE=0 TestServerListLLMProviders_WorkingFunnelEndToEnd (internal/server)
RED_MODE=1 (RED guards reproduce pre-fix invariant)

```

SP2 catalog tests — **5 PASS**:

```

RED_MODE=0 go test -v ./internal/catalog/...
--- PASS: TestCatalog_UnifiedList
--- PASS: TestCatalog_HonestEmptyWhenVerifierDisabled
--- PASS: TestCatalog_NamingGrammar
--- PASS: TestHandler_CatalogRoute_Polarity
--- PASS: TestHandler_CatalogEndpoint
ok dev.helix.agent/internal/catalog

```

2.3 §1.1 paired mutations — both FAIL-on-mutation (guards are genuine)

SP1 mutation — revert `listLLMModels` to `GetVerifiedModels(ctx)` (drop key gate):

```

RED_MODE=0 TestServerListLLMModels_WorkingFunnelEndToEnd
FAIL - "failed model must be hidden", "pending model must be hidden",
      "no-key provider model must be hidden", served map has 5 items (w

```

Restored; `handlers.go` diff back to original (19+/4-). Guard is NOT a tautology.

SP2 mutation — drop the aggregate ensemble entry from `Build()`:

```

RED_MODE=0 TestCatalog_UnifiedList
FAIL - "unified catalog missing required targets [ensemble]"

```

Restored; package tests GREEN again. Guard is NOT a tautology.

2.4 Regression — full touched-package suites PASS

```

go test ./internal/llm/ ./internal/server/ ./cmd/cli/ ./cmd/server/ (helix)
go test ./internal/catalog/... (helix)
go build ./internal/router/... (helix)

```

3. Anti-bluff assessment

- **Guards genuine** — both paired mutations flip GREEN→FAIL (§2.3). Not config-only / not tautologies.
- **No new bluff strings** — simulated | for now | TODO implement | in production this would scan of the SP1 + SP2 **production** files is clean. The one hit, `submodules/helix_agent/internal/router/router.go:458` (// For now, log that messaging...), is **pre-existing** (confirmed not in this diff) and out of scope; the messaging adapter is unrelated to the catalog block at :767–785.
- **Honest-empty correct** — `catalog.go Build()` emits a `KindModel` entry only when `verified != nil AND vm.Verified==true`;
`NewDiscoveryVerifiedSource(nil)` returns a nil source ⇒ no fabricated model

list. TestCatalog_HonestEmptyWhenVerifierDisabled proves it. The server funnel hides failed/pending/sub-threshold/no-key models (verified end-to-end over real httptest HTTP, not metadata-only).

- **CONST-042** — no API-key value is logged/printed in any changed production file; `loadAPIKeysAtStartup()` returns only a bool, comment explicitly states values are never logged.
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4. No-regression / dead-code / scope findings

- **No sibling break** — full suites of all five touched packages PASS (§2.4). `listLLMProviders + listLLMModels` both migrated symmetrically; `buildProvidersFromVerifiedModels` reused unchanged.
 - **No route collision (SP2)** — `/catalog` is registered exactly once (router.go:785). No pre-existing `/catalog` route. `os` already imported (router.go:47), so `os.Getenv("USE_HELIX_LLM")` compiles. Import reorder is cosmetic (gofmt-grouped), no functional change.
 - **No go.mod / go.sum edits** in either lane. Lanes are disjoint (no shared file).
 - **CLI-local presentProviders + providerEnvAliases are now production-dead** (finding F1, minor) — the production call site (`cmd/cli/main.go:1370`) switched to `llm.PresentProviderNames()`; the only remaining live references to `presentProviders/providerEnvAliases` are tests (`listmodels_d2_test.go, listmodels_funnel_source_test.go`). Per §11.4.124 the wave correctly did NOT delete on sight (extra-caution default), and the funnel-source test even uses the now-orphaned table as a *negative discriminator* (proving qwen recognition can only come from the committed table). Acceptable as-is; flagged for a future tracked §11.4.124 cleanup (investigate git-history + remove-or-rewire in a removal-only commit). Not a blocker.
-

5. Informational note (not a finding)

- **SP2 /catalog model section is latent at the wired endpoint** — router.go passes `Verified: NewDiscoveryVerifiedSource(nil)`, so the live endpoint will never surface `<provider>/<model>` entries until a `ModelDiscoveryService` is wired in `SetupRouterWithContext`. This is **consistent** with the surrounding code (the existing discovery path at router.go:1314 is also explicitly `nil` — the service genuinely isn't available in this scope) and is the **anti-bluff-correct** choice (honest-empty over fabrication). The model-join logic is fully implemented + unit-tested via the registry/discovery adapters; it activates the moment discovery is wired project-wide. No action required for this wave.
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6. Verdict

GO — both lanes build + vet clean, every new test passes at GREEN, both §1.1 paired mutations genuinely FAIL, no regression in full touched-package suites, anti-bluff posture sound (genuine guards, honest-empty, no new bluff strings, CONST-042 respected), lanes disjoint, no go.mod edits, no route collision.

Must-fix (ordered): none (zero blocking).

Recommended (non-blocking): 1. F1 — track a §11.4.124 follow-up to retire the now-dead CLI-local `presentProviders + providerEnvAliases` (`helix_code/cmd/cli/main.go:1430,1461`) in a removal-only commit after git-history investigation, and migrate/retire `cmd/cli/listmodels_d2_test.go`'s `presentProviders` assertions.

Captured build/test PASS-FAIL lines

```
SP1 build : go build ./internal/llm/... ./internal/server/... ./cmd/serve
SP1 vet   : go vet   (same set)
SP2 build : go build ./internal/catalog/... ./internal/router/...
SP2 vet   : go vet   ./internal/catalog/...
SP1 GREEN : 5/5 funnel+wiring tests RED_MODE=0
SP2 GREEN : 5/5 catalog tests
SP1 mut   : revert handler→GetVerifiedModels ⇒ TestServerListLLMModels_Wo
SP2 mut   : drop ensemble entry ⇒ TestCatalog_UnifiedList
Regression : full suites internal/llm internal/server cmd/cli cmd/server (
Regression : full suite internal/catalog (helix_agent)
```

Finding counts: blocking = 0 · non-blocking (recommended) = 1 (F1) · informational = 1.
VERDICT: GO.